决策指标	10万英镑预算	7万英镑预算
预算上限 GBP	100,000	70,000
推荐车辆	**货物线 G1：3辆**	**货物线 G1：2辆**
实际采购 GBP	84,000	56,000
预算余额 GBP	16,000	14,000
基准首年净现金 GBP	192,270	128,180
压力首年净现金 GBP	**137,010**	**91,340**
可行组合数	23	5
较10万方案节省采购 GBP	0	28,000
较10万方案少获压力净现金 GBP	0	45,670
压力净现金保留比例	100.0%	66.7%

选项	车数	情景	Base	距离	时间因子	批容量	总运量	满批数	尾批量	采购单价	年运行单价	满批收入	尾批收入	年收入	采购额	年运行费	首年净现金
C1		0 base		5916	200	234	120	0	0	0	22000	7000	15842	0	0	0	0	0
C1		0 stress		5916	200	230	120	0	0	0	22000	7000	15571	0	0	0	0	0
C1		1 base		5916	200	234	120	480	4	0	22000	7000	15842	0	63368	22000	7000	34368
C1		1 stress		5916	200	230	120	360	3	0	22000	7000	15571	0	46713	22000	7000	17713
C1		2 base		5916	200	234	120	960	8	0	22000	7000	15842	0	126736	44000	14000	68736
C1		2 stress		5916	200	230	120	720	6	0	22000	7000	15571	0	93426	44000	14000	35426
C1		3 base		5916	200	234	120	1200	10	0	22000	7000	15842	0	158420	66000	21000	71420
C1		3 stress		5916	200	230	120	960	8	0	22000	7000	15571	0	124568	66000	21000	37568
G1		0 base		6144	300	242	80	0	0	0	28000	10000	17015	0	0	0	0	0
G1		0 stress		6144	300	238	80	0	0	0	28000	10000	16734	0	0	0	0	0
G1		1 base		6144	300	242	80	480	6	0	28000	10000	17015	0	102090	28000	10000	64090
G1		1 stress		6144	300	238	80	400	5	0	28000	10000	16734	0	83670	28000	10000	45670
G1		2 base		6144	300	242	80	960	12	0	28000	10000	17015	0	204180	56000	20000	128180
G1		2 stress		6144	300	238	80	800	10	0	28000	10000	16734	0	167340	56000	20000	91340
G1		3 base		6144	300	242	80	1440	18	0	28000	10000	17015	0	306270	84000	30000	192270
G1		3 stress		6144	300	238	80	1200	15	0	28000	10000	16734	0	251010	84000	30000	137010
G2		0 base		6144	300	252	50	0	0	0	45000	18000	11074	0	0	0	0	0
G2		0 stress		6144	300	248	50	0	0	0	45000	18000	10898	0	0	0	0	0
G2		1 base		6144	300	252	50	450	9	0	45000	18000	11074	0	99666	45000	18000	36666
G2		1 stress		6144	300	248	50	350	7	0	45000	18000	10898	0	76286	45000	18000	13286
G2		2 base		6144	300	252	50	900	18	0	45000	18000	11074	0	199332	90000	36000	73332
G2		2 stress		6144	300	248	50	700	14	0	45000	18000	10898	0	152572	90000	36000	26572
G2		3 base		6144	300	252	50	1350	27	0	45000	18000	11074	0	298998	135000	54000	109998
G2		3 stress		6144	300	248	50	1050	21	0	45000	18000	10898	0	228858	135000	54000	39858
P1		0 base		3185	160	243	100	0	0	0	20000	8000	5904	0	0	0	0	0
P1		0 stress		3185	160	239	100	0	0	0	20000	8000	5807	0	0	0	0	0
P1		1 base		3185	160	243	100	700	7	0	20000	8000	5904	0	41328	20000	8000	13328
P1		1 stress		3185	160	239	100	600	6	0	20000	8000	5807	0	34842	20000	8000	6842
P1		2 base		3185	160	243	100	1400	14	0	20000	8000	5904	0	82656	40000	16000	26656
P1		2 stress		3185	160	239	100	1200	12	0	20000	8000	5807	0	69684	40000	16000	13684
P1		3 base		3185	160	243	100	2050	20	50	20000	8000	5904	2952	121032	60000	24000	37032
P1		3 stress		3185	160	239	100	1640	16	40	20000	8000	5807	2323	95235	60000	24000	11235
P2		0 base		3185	160	249	70	0	0	0	30000	12000	4235	0	0	0	0	0
P2		0 stress		3185	160	245	70	0	0	0	30000	12000	4167	0	0	0	0	0
P2		1 base		3185	160	249	70	840	12	0	30000	12000	4235	0	50820	30000	12000	8820
P2		1 stress		3185	160	245	70	630	9	0	30000	12000	4167	0	37503	30000	12000	-4497
P2		2 base		3185	160	249	70	1680	24	0	30000	12000	4235	0	101640	60000	24000	17640
P2		2 stress		3185	160	245	70	1260	18	0	30000	12000	4167	0	75006	60000	24000	-8994
P2		3 base		3185	160	249	70	2050	29	20	30000	12000	4235	1210	124025	90000	36000	-1975
P2		3 stress		3185	160	245	70	1640	23	30	30000	12000	4167	1786	97627	90000	36000	-28373
M1		0 base		4550	240	251	60	0	0	0	25000	6000	7841	0	0	0	0	0
M1		0 stress		4550	240	247	60	0	0	0	25000	6000	7716	0	0	0	0	0
M1		1 base		4550	240	251	60	240	4	0	25000	6000	7841	0	31364	25000	6000	364
M1		1 stress		4550	240	247	60	180	3	0	25000	6000	7716	0	23148	25000	6000	-7852
M1		2 base		4550	240	251	60	480	8	0	25000	6000	7841	0	62728	50000	12000	728
M1		2 stress		4550	240	247	60	360	6	0	25000	6000	7716	0	46296	50000	12000	-15704
M1		3 base		4550	240	251	60	677	11	17	25000	6000	7841	2221	88472	75000	18000	-4528
M1		3 stress		4550	240	247	60	540	9	0	25000	6000	7716	0	69444	75000	18000	-23556

预算案例	排名	车辆组合	采购额 GBP	基准净现金 GBP	压力净现金 GBP	压力收益距最优 GBP
10万预算	1	货物线 G1：3辆	84000	192270	137010	0
10万预算	2	煤炭线 C1：2辆；货物线 G1：2辆	100000	196916	126766	10244
10万预算	3	煤炭线 C1：1辆；货物线 G1：2辆；客运线 P1：1辆	98000	175876	115895	21115
10万预算	4	煤炭线 C1：1辆；货物线 G1：2辆	78000	162548	109053	27957
7万预算	1	货物线 G1：2辆	56000	128180	91340	0
7万预算	2	煤炭线 C1：1辆；货物线 G1：1辆；客运线 P1：1辆	70000	111786	70225	21115
7万预算	3	煤炭线 C1：1辆；货物线 G1：1辆	50000	98458	63383	27957
7万预算	4	货物线 G1：1辆；客运线 P1：2辆	68000	90746	59354	31986

输入项	值	来源
year_days	360	planning_policy.json
fleet_limit	4	planning_policy.json
stress_demand_percent	80	planning_policy.json
stress_age_periods	4	planning_policy.json
stress_cycle_days	10	planning_policy.json
minimum_stress_net	50000	planning_policy.json
max_per_option	3	planning_policy.json
currency	"GBP"	planning_policy.json
budget_cases	[{"id": "standard", "budget": 100000}, {"id": "reduced",	planning_policy.json
C1.id	"C1"	routes.json
C1.cargo	"Coal"	routes.json
C1.from	[20, 20]	routes.json
C1.to	[160, 80]	routes.json
C1.travelled_tiles	280	routes.json
C1.station_wait_days	12	routes.json
C1.age_periods	28	routes.json
C1.cycle_days	90	routes.json
C1.capacity	120	routes.json
C1.demand	1200	routes.json
C1.capex	22000	routes.json
C1.annual_running	7000	routes.json
C1.exclusive_group	"coal"	routes.json
G1.id	"G1"	routes.json
G1.cargo	"Goods"	routes.json
G1.from	[40, 60]	routes.json
G1.to	[240, 160]	routes.json
G1.travelled_tiles	420	routes.json
G1.station_wait_days	4	routes.json
G1.age_periods	18	routes.json
G1.cycle_days	60	routes.json
G1.capacity	80	routes.json
G1.demand	1600	routes.json
G1.capex	28000	routes.json
G1.annual_running	10000	routes.json
G1.exclusive_group	"goods"	routes.json
G2.id	"G2"	routes.json
G2.cargo	"Goods"	routes.json
G2.from	[40, 60]	routes.json
G2.to	[240, 160]	routes.json
G2.travelled_tiles	340	routes.json
G2.station_wait_days	8	routes.json
G2.age_periods	8	routes.json
G2.cycle_days	40	routes.json
G2.capacity	50	routes.json
G2.demand	1600	routes.json

输入项	值	来源
G2.capex	45000	routes.json
G2.annual_running	18000	routes.json
G2.exclusive_group	"goods"	routes.json
P1.id	"P1"	routes.json
P1.cargo	"Passengers"	routes.json
P1.from	[15, 15]	routes.json
P1.to	[115, 75]	routes.json
P1.travelled_tiles	240	routes.json
P1.station_wait_days	6	routes.json
P1.age_periods	12	routes.json
P1.cycle_days	48	routes.json
P1.capacity	100	routes.json
P1.demand	2050	routes.json
P1.capex	20000	routes.json
P1.annual_running	8000	routes.json
P1.exclusive_group	"passengers"	routes.json
P2.id	"P2"	routes.json
P2.cargo	"Passengers"	routes.json
P2.from	[15, 15]	routes.json
P2.to	[115, 75]	routes.json
P2.travelled_tiles	180	routes.json
P2.station_wait_days	4	routes.json
P2.age_periods	6	routes.json
P2.cycle_days	30	routes.json
P2.capacity	70	routes.json
P2.demand	2050	routes.json
P2.capex	30000	routes.json
P2.annual_running	12000	routes.json
P2.exclusive_group	"passengers"	routes.json
M1.id	"M1"	routes.json
M1.cargo	"Mail"	routes.json
M1.from	[80, 10]	routes.json
M1.to	[200, 130]	routes.json
M1.travelled_tiles	310	routes.json
M1.station_wait_days	12	routes.json
M1.age_periods	24	routes.json
M1.cycle_days	90	routes.json
M1.capacity	60	routes.json
M1.demand	677	routes.json
M1.capex	25000	routes.json
M1.annual_running	6000	routes.json
M1.exclusive_group	"mail"	routes.json